

Multiple Choice Exam: Computational Social Science with Images and Audio

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Instructions

Select the best answer for each question. For each question, only one answer is correct. The exam lasts 60 minutes and comprises 41 questions, but one question is bonus (you can choose one question you want to skip, or you can solve all questions and get a bonus point). In other words, to get the maximum of points, you will need to solve **40 questions**, yielding an average of **1.5 minutes per question**. This exam is closed-book.

It was a great pleasure having this class with all of you. We greatly enjoyed your presentations and your insights! We wish you the best of luck, in this exam and in your studies in general :)

Question 1. What does David Marr's definition of computer vision emphasize?

- A. Automating human visual processes
- B. Inverting image formation
- C. Information processing from raw data
- D. The inverse of computer graphics

Question 2. In a grayscale image represented as a two-dimensional function $f(x, y)$, what does each point represent?

- A. A vector of three color channels
- B. A frequency component
- C. A pixel with an intensity value
- D. It depends on the context

Question 3. In standard RGB (sRGB) representation, how many distinct colors can be represented?

- A. Approximately 1 million
- B. Approximately 16.7 million
- C. Approximately 281 trillion
- D. Unlimited colors

Question 4. In the study on portrayals of race and gender in children's literature, which pre-trained model was fine-tuned for sentiment analysis?

- A. BERT
- B. GPT-2
- C. ResNet
- D. CLIP

Question 5. What is the main computational challenge when working with unstructured data that makes common traditional estimation algorithms unsuitable?

- A. Large number of observations (n), high dimensionality (p), and often multiple modalities
- B. The data is too noisy for traditional methods

- C. Unstructured data cannot be converted to numerical format
- D. The data requires manual labeling before processing

Question 6. What was the significance of AlexNet winning the ImageNet Challenge in 2012?

- A. It was the first use of hand-crafted features in image classification
- B. It introduced the transformer architecture to computer vision
- C. It demonstrated that CPUs are superior to GPUs for training
- D. It was the first time a deep CNN outperformed traditional machine learning methods on large-scale data

Question 7. Which of the following was NOT a key innovation introduced by AlexNet?

- A. Use of GPUs for training
- B. ReLU activation functions
- C. Self-attention mechanisms
- D. Dropout regularization

Question 8. In the Generalized Topic Model (GTM), what technique is used to approximate the intractable marginal likelihood?

- A. Markov Chain Monte Carlo (MCMC) sampling
- B. Expectation-Maximization (EM) algorithm
- C. Regularized Variational Autoencoder (VAE)
- D. Gibbs sampling with collapsed inference

Question 9. In the study on aligning AI with public values through deliberation, what voting mechanism increases the cost of votes quadratically (e.g., 2 votes cost 4 tokens, 3 votes cost 9 tokens)?

- A. Weighted voting
- B. Plurality voting
- C. Quadratic voting
- D. Ranked-choice voting

Question 10. What is the key innovation of the Vision Transformer (ViT)?

- A. Treating images as sequences of patches
- B. Using deeper convolutional layers
- C. Applying recurrent connections between layers
- D. Using hand-crafted edge detection filters

Question 11. In the context of CNNs, what do layers close to the input typically learn?

- A. High-level semantic features
- B. Classification probabilities
- C. Complete object representations
- D. Low-level features such as edges and lines

Question 12. What is the purpose of Maximum Mean Discrepancy (MMD) in the GTM loss function?

- A. To maximize the reconstruction accuracy
- B. To improve prediction of auxiliary outcomes
- C. To reduce overfitting in the decoder network
- D. To keep the encoder distribution close to the prior distribution

Question 13. What does a color histogram represent?

- A. The spatial arrangement of colors in an image
- B. The distribution of pixel intensity levels
- C. The gradient orientations in an image
- D. The edges detected in an image

Question 14. What does high entropy in an image region indicate about that region?

- A. A more complex texture with more details
- B. A more uniform texture with fewer details
- C. The presence of faces in the image
- D. Strong horizontal edges

Question 15. What do Histogram of Oriented Gradients (HOG) features capture?

- A. Color distribution across the image
- B. Audio frequency components
- C. The distribution and intensity of gradients in an image
- D. Pixel brightness only

Question 16. According to Beraja, Yang, and Yuchtman (2023, discussed in the second part of the first lecture), what is the key question their research addresses?

- A. Whether deep learning can replace classical machine learning
- B. Whether CNNs outperform transformers
- C. Whether audio analysis is more important than image analysis
- D. Whether access to government data stimulates commercial AI innovation

Question 17. For the urban visual intelligence study (i.e., the study on describing city profiles), we discussed that Street View features ...

- A. Estimate travel behaviors particularly well
- B. Are a great estimator of certain health outcomes
- C. Are not limited by the coverage of Google Street View
- D. Do not outperform points of interest (POI)

Question 18. We discussed a study on the relationship between political unrest and AI procurement in China. What did these authors find?

- A. Unrest decreases AI procurement
- B. Unrest drives increased AI procurement by public security agencies
- C. Unrest has no effect on AI procurement
- D. AI procurement causes political unrest

Question 19. We discussed a study on the limits of AI's effectiveness for authoritarian control (Yang, 2024). Which limit did we discuss in this context?

- A. Strategic behavior that corrupts training data
- B. Privacy protection legislation that prevents the use of certain data
- C. The generally low usefulness of AI for authoritarian control
- D. AI only works in democratic societies

Question 20. In the study on racial avoidance on New York City streets, what outcome measure was used to quantify pedestrian avoidance behavior?

- A. Walking speed differential
- B. Eye contact duration
- C. Standardized pedestrian deviation

D. Time to cross intersection

Question 21. In the macro nowcasting application using GTM, what was the prediction target when analyzing *Wall Street Journal* articles?

- A. Stock market returns
- B. Unemployment rate
- C. Log Industrial Production Growth
- D. Consumer confidence index

Question 22. In the study on visual representation and stereotypes in news media, which ethnic group was found to be disproportionately associated with images in articles about crime and poverty?

- A. Asians
- B. Hispanics
- C. Blacks
- D. Whites

Question 23. In Cantu (2019), what method was used to detect election fraud in Mexico's 1988 presidential election?

- A. Analyzing original vote-tally sheet images to detect alterations
- B. Comparing exit poll data with official results
- C. Using audio recordings of vote counting
- D. Surveying voters after the election

Question 24. According to the lectures, what is the relationship between CNNs and feature learning compared to classical machine learning?

- A. Both require explicit hand-crafted feature engineering

- B. Classical ML learns features automatically while CNNs require manual features
- C. CNNs and classical ML use identical feature extraction approaches
- D. CNNs learn features automatically while classical ML typically requires explicit feature extraction

Question 25. In the study on AI and political unrest, what instrumental variable was used to address the endogeneity of political unrest?

- A. Historical protest data
- B. Local weather conditions
- C. Economic indicators
- D. Social media activity

Question 26. In the context of sound waves, what does wavelength determine?

- A. The loudness of the sound
- B. The pitch of the sound
- C. The duration of the sound
- D. The timbre of the sound

Question 27. What is the typical human hearing frequency range?

- A. 1 Hz to 100 Hz
- B. 100 Hz to 100,000 Hz
- C. 20 Hz to 20,000 Hz
- D. 1,000 Hz to 50,000 Hz

Question 28. What happens to pitch when you double the frequency of a sound?

- A. It decreases by one octave
- B. It stays the same
- C. It becomes inaudible
- D. It increases by one octave

Question 29. What embedding model can one use to handle both text and images?

- A. BERT
- B. Word2Vec
- C. GloVe
- D. LongCLIP

Question 30. What does the sampling frequency in digital audio represent?

- A. The loudness of the audio
- B. The frequency of the sound being recorded
- C. The bit depth of the recording
- D. The number of samples taken per second from a continuous signal

Question 31. What is the standard CD sampling rate?

- A. 22.05 kHz
- B. 96 kHz
- C. 44.1 kHz
- D. 192 kHz

Question 32. In audio signal processing, what does amplitude represent?

- A. The frequency of the sound

- B. The magnitude of change in the pressure wave from equilibrium
- C. The duration of the sound
- D. The number of channels

Question 33. What is the Mel scale designed to reflect?

- A. Linear frequency perception
- B. Wavelength of sound
- C. Amplitude of sound
- D. Human perception of pitch

Question 34. Why is audio typically divided into short frames (e.g., 20-40 ms) for MFCC extraction?

- A. To increase file size
- B. To remove noise
- C. To allow assuming stationarity within each frame
- D. To increase the sampling rate

Question 35. What is the primary advantage of using spectrograms over raw audio waveforms as input for transformer-based speech recognition models?

- A. Spectrograms contain more information than raw audio
- B. Spectrograms are easier to label manually
- C. Spectrograms require no preprocessing
- D. Spectrograms significantly reduce sequence length while preserving relevant information

Question 36. What does the Fourier Transform do to a time-domain signal?

- A. Compresses it

- B. Decomposes it into constituent frequencies
- C. Amplifies it
- D. Removes noise

Question 37. What type of input does OpenAI’s Whisper speech recognition model use?

- A. Raw audio waveforms only
- B. MFCCs only
- C. Binary audio data
- D. Log-mel spectrograms

Question 38. What distinguishes “color gamut” from “color depth”?

- A. Gamut refers to the range/variety of colors; depth refers to the number of discrete values
- B. They are the same thing
- C. Gamut refers to brightness; depth refers to saturation
- D. Gamut is for images; depth is for audio

Question 39. According to the lectures, why do we (sometimes) still care about explicit feature extraction in 2025?

- A. Deep learning has been abandoned
- B. Classical ML is superior in most cases
- C. For interpretability and computational constraints
- D. Many deep learning architectures still require explicit feature extraction

Question 40. In the Salazar-Miranda et al. study on urban spaces, they found that, between the 1980s and 2010s, ...

- A. Measuring whether people moved in groups was impossible methodologically speaking
- B. People lingering in public spaces stayed the same once they corrected for measurement changes over time
- C. People lingering in public spaces decreased
- D. People moving in groups increased

Question 41. In the study on economic development through paintings, what neural network architecture was used to predict emotions from artwork?

- A. ResNet pre-trained on COCO
- B. VGG-16 pre-trained on Places365
- C. EfficientNet pre-trained on ImageNet
- D. BERT fine-tuned on art descriptions